



PIONEER JUNIOR COLLEGE, SINGAPORE

JC2 PRELIMINARY EXAMINATION
HIGHER 2

CHEMISTRY

9746/03

Paper 3 Free-Response

17 September 2009

2 hours

Additional Materials: Answer Paper
 Data Booklet
 Cover Page

READ THESE INSTRUCTIONS FIRST

Write your name, CT Group and index number in the spaces at the top of this page and every sheet of your scripts.

Write in dark blue or black pen.

Choose **4** out of 5 Questions.

Write your answers on the writing papers provided. **Begin** each question on a **fresh** page. If there is no answer to any of the questions, a blank sheet of paper for each unanswered question with the question number(s) must still be submitted.

A Data Booklet is provided.

You may use a calculator.

At the end of the examination, fasten all your work securely together.

The number of marks is given in brackets [] at the end of each question or part question.

This document consists of **8** printed pages.

- 1 Iron(III) bromide is a fairly strong Lewis acid, and is used as a catalyst in organic synthesis. Iron(III) bromide dissolves in water to give a brown, acidic and corrosive solution.

(a) When 0.754 g of FeBr_3 is dissolved in 250 cm^3 of water, the resulting solution is found to have a pH of 3.6.

- (i) Explain, with the aid of balanced equation(s), why a solution of iron(III) bromide has pH less than 7.
- (ii) Using your equation(s) in (i), calculate the K_a of $\text{Fe}^{3+}(\text{aq})$.

[4]

(b) (i) Define the term *standard enthalpy change of formation* with respect to iron(III) bromide.

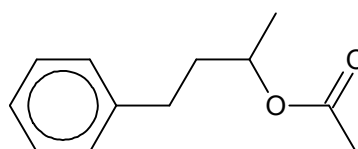
(ii) By using relevant data from the *Data Booklet* and the following data, draw an energy level diagram and calculate the lattice energy of iron(III) bromide.

First electron affinity of bromine	-325 kJ mol^{-1}
Enthalpy change of atomisation of bromine	$+112 \text{ kJ mol}^{-1}$
Enthalpy change of atomisation of iron	$+350 \text{ kJ mol}^{-1}$
Enthalpy change of formation of iron(III) bromide.	-320 kJ mol^{-1}

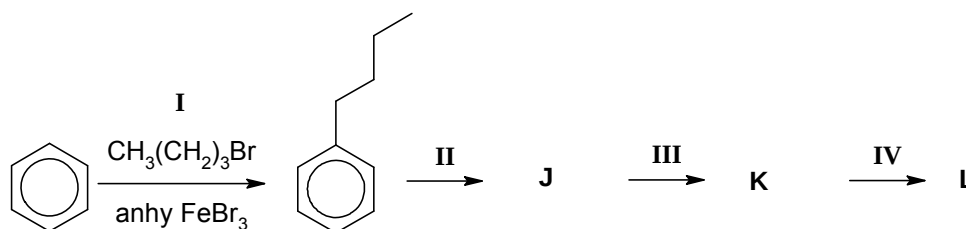
(iii) The experimental value for the lattice energy of iron(III) bromide is found to be $-5268 \text{ kJ mol}^{-1}$. Comment on and suggest reasons for the difference between this value and the value you obtained in (b) (ii).

[7]

(c) Benzene reacts with 1-bromobutane to form butylbenzene in the presence of anhydrous catalyst, FeBr_3 to produce compound L, in the following reaction scheme

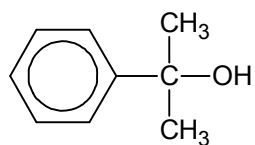


Compound L



- (i) Name the mechanism of the reaction in Stage I and hence explain the need for FeBr_3 to be anhydrous.
- (ii) Suggest the identities of J and K.

- (iii) State the reagents and conditions for stages II to IV.
- (iv) Suggest how you can differentiate between butylbenzene and Compound M, as shown below.



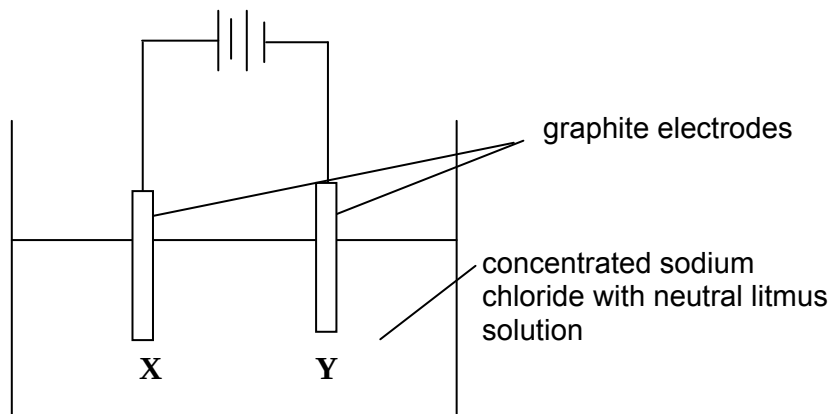
Compound M

[9]

[Total: 20]

2 This question is about the important properties of chlorine and its compounds.

(a) The diagram shows the electrolysis of a concentrated solution of sodium chloride solution.



- (i) By quoting and using relevant $E^\ominus$ values from the *Data Booklet*, state and account for the colours you will expect to see at regions X and Y when the switch is closed.
- (ii) Sodium chlorate(V) can be prepared when the products of the above electrolysis of concentrated sodium chloride are mixed. With the aid of a balanced equation, explain how sodium chlorate(V) can be formed.
- (iii) Using the equation in (a)(ii), calculate the concentration of NaClO_3 formed when a current of 5 A is passed through a 2.0 dm^3 electrolyte containing concentrated sodium chloride solution for 10 hours.

[9]

(b) Chlorine is an important chemical for water treatment. However, chlorine dioxide is now commonly used due to its greater solubility in water compared to chlorine.

- (i) Draw the dot-and-cross diagram for chlorine dioxide and suggest a reason why chlorine dioxide is a good oxidising agent.
- (ii) Explain why chlorine dioxide is relatively more soluble in water compared to chlorine.
- (iii) Chlorine dioxide can be made in the laboratory by reacting silver chlorate(V) with chlorine. The reaction also gives a white precipitate and a colourless gas.

Identify the white precipitate and the colourless gas and write an equation for the reaction between silver chlorate(V) and chlorine.

[5]

(c) Chromium reacts with chlorine to form CrCl_3 . However, when made to react with other halogens such as fluorine and iodine, the main products of the reaction are CrF_4 and CrI_2 respectively.

Using relevant data from the *Data Booklet*, explain the above observations, in terms of the relative oxidising powers of Group VII elements.

[2]

- (d) A metallic salt containing the anion $I_xCl_y^-$ decomposed when heated to form iodine vapour and the metallic chloride. When 0.270 g of this compound was heated, the iodine evolved was bubbled into excess potassium iodide solution.

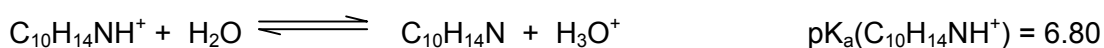
The dissolved iodine required 21.30 cm^3 of $0.0600\text{ mol dm}^{-3}$ aqueous sodium thiosulfate for complete reaction. When the residue was dissolved in water, the resulting solution required 6.40 cm^3 of 0.100 mol dm^{-3} aqueous silver nitrate for complete reaction.

- (i) Determine the value of x and y in $I_xCl_y^-$.
- (ii) Hence determine the identity of the metallic element.

[4]

[Total:20]

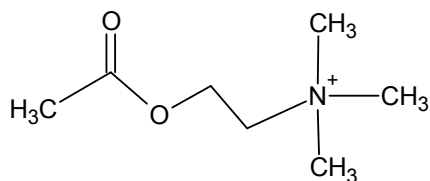
- 3 Nicotine ($C_{10}H_{14}N$) is a drug of addiction present in tobacco. In aqueous solution, both nicotine and its conjugate acid exist in equilibrium as shown below.



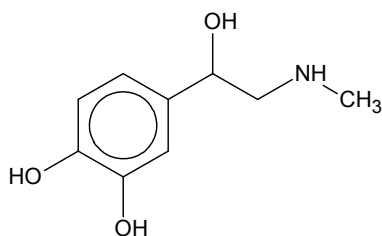
- (a) (i) Calculate the pH of a $0.100 \text{ mol dm}^{-3} C_{10}H_{14}NH^+$ at 25°C .
- (ii) Nicotine and its conjugate acid are present in the fatty tissue of the bladder walls and in the urine inside the bladder. If the pH of the urine decreases, explain what happens to the concentration of the conjugate acid of nicotine in the urine of a smoker's body.
- (iii) Calculate the pH of a 5.00 dm^3 solution consisting of $0.100 \text{ mol dm}^{-3}$ nicotine, $C_{10}H_{14}N$ with $0.200 \text{ mol dm}^{-3}$ of its conjugate acid, $C_{10}H_{14}NH^+$.
- (iv) Calculate what volume of $0.100 \text{ mol dm}^{-3}$ of H_2SO_4 (aq) needs to be added to the solution in (a)(iii) to obtain a buffer solution of pH = 6.40.

[10]

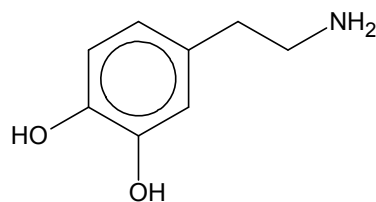
When a cigarette is smoked, nicotine-rich blood stimulates the release of many chemical messengers including *acetylcholine*, *epinephrine* and *dopamine*.



Acetylcholine



Epinephrine



Dopamine

- (b) (i) Name the type of isomerism exhibited by *Epinephrine* and draw the isomers
- (ii) State and explain which compound, *Epinephrine* or *Dopamine* is a stronger base.
- (c) (i) A reaction between *Dopamine* and chloromethane, CH_3Cl forms a compound with formula $C_{10}H_{15}NO_2$. Suggest a structure for this compound and suggest how the yield of this compound can be **maximised**.
- (ii) Describe using only two chemical tests, how you are able to distinguish the three chemical messengers: Dopamine, Epinephrine and Acetylcholine

[5]

[Total: 20]

- 4 This question is about organic compounds **P** and **Q**, which have the same molecular formula.

P and **Q** have a relative molecular mass of 118 and contain 50.8% C, 8.5% H and 40.7% O. Both **P** and **Q** are optically active.

P reacts readily with cold aqueous sodium hydroxide but **Q** only reacts with hot aqueous sodium hydroxide to give a salt **R**, and **S**. **P** reacts with PCl_5 to give misty fume but does not react with hot acidified potassium dichromate.

When **Q** is heated with acidified potassium manganate, effervescence is observed and **T** is formed. One mole each of **S** and **T** reacts with excess aqueous alkaline iodine to give two mole of a yellow precipitate.

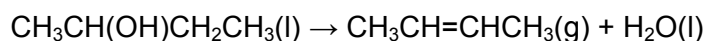
- (a) (i) Calculate the molecular formula of **P** and **Q**.
 (ii) Deduce the structure of **P** to **T**. Explain the chemistry of the reactions.

[11]

- (b) Outline the synthesis of **P** from butan-2-ol in the laboratory.

[3]

- (c) Butan-2-ol can be dehydrated to give the major product of trans-but-2-ene.



- (i) Suggest why the trans-isomer of but-2-ene is formed in a greater proportion compared to the cis-isomer.
 (ii) The standard entropy change of the reaction is $+154 \text{ J K}^{-1} \text{ mol}^{-1}$. Explain why the sign is positive.
 (iii) Using the value in (ii), and the following values of standard enthalpy changes of combustion, calculate the standard Gibbs free energy of the reaction.

species	$\Delta H_c^\theta / \text{kJ mol}^{-1}$
$\text{CH}_3\text{CH}(\text{OH})\text{CH}_2\text{CH}_3(\text{l})$	- 2660
trans- $\text{CH}_3\text{CH}=\text{CHCH}_3(\text{g})$	- 2710

- (iv) Use your answer in (iii) to explain the following statement.

"The dehydration of butan-2-ol to but-2-ene takes place when heated and not at room temperature."

[6]

[Total: 20]

- 5 (a) Oxygenated human blood is red in colour due to the presence of conjugated proteins known as haemoglobin. Haemoglobin consists of oxygen-carrying organic molecules that surround a *transition element* ion.

- (i) Define the term *transition element* and name the *transition element* that is present in haemoglobin.
- (ii) Explain, using the crystal field theory, how the red colour arises in human blood.

[5]

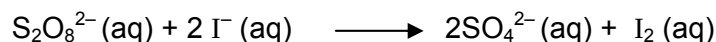
- (b) Vanadium is one of the many transition metals that are of great industrial importance. Other than its use to manufacture alloys, many of its compounds contain complexes that can function to speed up chemical reactions.

The dioxovanadium(V) ion, VO_2^+ can be added to a reaction mixture containing iodide ions, I^- and peroxodisulfate ions, $\text{S}_2\text{O}_8^{2-}$ to increase the rate of reaction.

- (i) State the role of the VO_2^+ in the reaction. Hence, explain, with the aid of a suitable diagram, how VO_2^+ increases the reaction rate.
- (ii) By considering suitable E^θ values in the *Data Booklet* and writing appropriate equations, show how the VO_2^+ functions in the reaction involving I^- and $\text{S}_2\text{O}_8^{2-}$.
- (iii) Sketch, on the same diagram, the two energy profiles of the reaction between I^- and $\text{S}_2\text{O}_8^{2-}$, with and without the addition of $\text{VO}_2^+(\text{aq})$.

[8]

- (c) The kinetics of the uncatalysed reaction between iodide ions, I^- and peroxodisulfate ions, $\text{S}_2\text{O}_8^{2-}$ was studied.



A continuous method with sampling was used to determine the rate equation. The procedure was as follows:

50 cm³ of $\text{K}_2\text{S}_2\text{O}_8(\text{aq})$ was mixed with 50 cm³ of $\text{KI}(\text{aq})$. Both solutions were of known concentrations. 10 cm³ of the reaction mixture was titrated against standard solution of $\text{Na}_2\text{S}_2\text{O}_3(\text{aq})$ using starch as an indicator. The table below shows the titrimetric results obtained from the experiment.

Time / min	4	8	12	16	20	∞
Volume of $\text{Na}_2\text{S}_2\text{O}_3(\text{aq})$ / cm ³	17.1	26.8	32.5	35.7	37.5	40

- (i) Give a reason why potassium peroxodisulfate, $\text{K}_2\text{S}_2\text{O}_8$ was used in large excess in the experiment.
- (ii) By plotting a graph or otherwise, determine the order of reaction with respect to iodide ions.
- (iii) The order of reaction with respect to $\text{S}_2\text{O}_8^{2-}$ is one. Explain how this may be determined experimentally.

[7]

[Total 20]